# Household Vehicle Purchase Timing: Utility, Research Questions, and Model Components

Research Proposal Within the Auto-Loan Research Exercise

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**Abstract**

This draft develops the household vehicle-purchase decision that precedes auto-loan demand. It organizes the decision into Need, Ability, and Choice, connects these questions to a provisional dynamic utility formulation, and defines purchase probabilities over six- and twelve-month horizons. Section 4 states which components of the formulation come from which papers, what we keep, and what we change, citing each paper's technical analysis in the companion review's appendix. Creditworthiness, the financing choice, and household risk attitude are left out at this stage; the data-generating process, identification, estimation, and empirical validation remain open.

## 1 Purpose and Scope

The broader auto-loan exercise asks which existing bank customers may take an auto loan over a future six- or twelve-month interval. This proposal develops the preceding household decision: whether and when to purchase a vehicle. A vehicle purchase, borrowing to fund that purchase, and originating a loan with Chase are different events. The present formulation concerns the first. Financing conditions enter because they affect what a household can afford; lender selection, offer design, and repayment-risk modeling remain outside this stage.

The objective is to connect a household's purchase probability to the utility of its transportation options and the resources needed to obtain them. Need, Ability, and Choice organize the economic questions within that decision. Section 2 sets out the utility argument; Section 3 states the research questions, a provisional mathematical formulation, and their hierarchy; Section 4 states which components of the formulation come from which papers, what we keep, and what we change. The papers' mathematics and assumptions are walked through in the companion literature review, cited here by section and step number. This is a provisional formulation for discussion. It reports no estimated parameters, identification proof, simulation results, or validated forecasts.

## 2 Household Utility and the Value of Buying Now

A household obtains transportation services from the vehicles it holds and from alternatives such as public transportation. A purchase changes those services, uses funds that could support other consumption, and changes future ownership costs. The relevant comparison is therefore the expected discounted utility of buying now against that of continuing with the current arrangement, repairing it, changing travel mode, or waiting. Random utility provides the choice-based foundation; dynamic

replacement models make the continuation value explicit (McFadden, 1974; Rust, 1987; Schiraldi, 2011).

### 2.1 What the household values

Let $h$ index households and $r$ index decision months. Write $s_{hr}$ for the household's pre-decision state and $d$ for a discrete action, including the vehicle involved when a transaction occurs. Let $\mathcal{X}_{hr}^d$ be its fleet immediately after action $d$, $c$ its nondurable consumption during the month, $m_{hr}$ its composition, and $\ell_{hr}$ its location and transport environment. A candidate flow utility is

$$u_h(s_{hr}, d, c) = \theta_h \log(c/\bar{c}) + g_h(\mathcal{X}_{hr}^d, m_{hr}, \ell_{hr}; d) - \kappa_h(s_{hr}, d), \qquad \theta_h > 0. \tag{1}$$

Here $\bar{c} > 0$ is a fixed reference consumption level, $g_h$ values transportation services, and $\kappa_h$ represents the nonmonetary inconvenience of adjustment, search, or switching. Monetary transaction costs belong in the budget. Allowing $g_h$ to depend on the action also accommodates changes in non-car travel arrangements. The logarithm is a tractable concave specification to investigate, rather than a functional form established by the cited studies.

The earlier working formulation valued successive vehicles separately and penalized fleet age. One possible version of its transport term is

$$g_h(\mathcal{X}^d, m, \ell; d) = \sum_{k=1}^{n^d} \gamma_{hk}(m, \ell) + \sum_{v \in \mathcal{X}^d} (b_h^\top x_v - \eta_h a_v) + \psi_h(m, \ell; d), \tag{2}$$

where $n^d$ is fleet size, $x_v$ records vehicle attributes other than age, $a_v$ is vehicle age, and $\psi_h$ captures the value of available non-car transport under the action. Separate $\gamma_{hk}$ terms allow the value of the first vehicle to differ from that of an additional vehicle. Household vehicle-holding studies motivate attention to fleet size and composition (Berkovec and Rust, 1985; Bhat and Pulugurta, 1998); they do not establish this additive specification. A positive age penalty is a candidate restriction to test, since age alone does not measure reliability or maintenance costs. Retaining vehicle-level information also avoids treating two different fleets with the same oldest vehicle age as identical.

### 2.2 How Need, Ability, and Choice enter

**Need** concerns the transportation services the household wants relative to those its current arrangements provide. Fleet deterioration, changing drivers, a different commute, and discretionary upgrading can change the gain from adjustment. In (1), these changes affect the transport term and the future states the household expects. Need does not require a separately observed emergency or life event before a purchase is possible.

**Ability** determines which actions and consumption plans satisfy the household's resource constraints. A particular vehicle can be infeasible while a cheaper vehicle remains attainable. Liquidity and borrowing terms therefore shape the feasible set and the consumption forgone when buying (Attanasio et al., 2008; Adams et al., 2009). Ability is not identified with the coefficient on consumption, nor with a universal credit-approval indicator.

**Choice** is the comparison of the total values of feasible actions. It includes the tradeoff between transport services and other consumption, adjustment costs, and the value of waiting. It is not represented by the transaction-cost term alone. These three layers describe different aspects of one decision; they need not be independent, separately observed, or chronological.

### 2.3 Why timing belongs in utility

Let $\mathcal{D}(s)$ be the feasible discrete actions and $\mathcal{C}(s,d)$ the feasible consumption choices under action $d$. The systematic value of an action is

$$v_h(s,d) = \max_{c \in \mathcal{C}(s,d)} \left\{ u_h(s,d,c) + \beta_h \mathbb{E}\left[ \bar{V}_h(s') \mid s,d,c \right] \right\}, \qquad 0 < \beta_h < 1, \tag{3}$$

where $\beta_h$ discounts future utility and $\bar{V}_h$ is the expected value before the next choice shocks are realized. With an additive action-specific shock $\varepsilon_{hr}(d)$, the household selects the action maximizing $v_h(s,d) + \varepsilon_{hr}(d)$. Thus buying can become attractive because current transportation improves, because the household's budget changes, or because waiting becomes less valuable. Anticipated events affect this comparison through expectations, not through future observations supplied to a forecast.

Equations (1)–(3) state a candidate decision problem. Their empirical interpretation requires a defensible state, budget, transition law, and shock distribution. Utility differences are measured on a normalized utility scale; they are not automatically monetary welfare gains. Counterfactual effects of changing prices or financing terms require additional identification and stability assumptions (Magnac and Thesmar, 2002). The next section makes these requirements and the questions they serve explicit.

## 3 Research Questions and Provisional Formulation

### 3.1 The outcome to explain and predict

The main question is: *How do a household's transportation requirements, financial opportunities, and intertemporal choices determine its probability of purchasing a vehicle over a specified future interval?* At the end of month $t$, define

$$F_h(t,H) = \mathbb{P}\left( T_h^P \le t + H \mid \mathcal{I}_{ht} \right), \qquad H \in \{6, 12\}, \tag{4}$$

where $T_h^P > t$ is the month of the next purchase and $\mathcal{I}_{ht}$ contains only information available at the forecast origin. The existing 24-month horizon can be retained as an extension; the primary horizons are six and twelve months. "Now" refers to the next decision month, corresponding to $H = 1$.

For this draft, a purchase includes a fleet addition or a replacement bought with cash or credit. Disposal without replacement and taking a new lease do not themselves count as purchases. They can change the household's state and its subsequent purchase probability. This event definition is a working choice for review: if the intended endpoint includes leased vehicles, the event set and labels must be broadened consistently. A household may buy after first disposing of a vehicle, and that later purchase still counts in (4).

The household is the analytical unit because the model concerns shared vehicle holdings and resources. Constructing that unit requires a verified household linkage. Complete fleet records, outside-bank balances, consumption, and financing opportunities are data requirements, not assumed properties of available bank data.

### 3.2 Questions within the three layers

**RQ1: Need—what changes the value of the current transportation arrangement?** The first subquestion concerns the starting position: the number, age, condition, and use of current

vehicles, together with other transport options. The second concerns changes in requirements: new drivers, household composition, work, residence, and unexpected loss or failure. The third concerns timing: whether adjustment pressure is gradual, anticipated, or sudden. The empirical task is to determine which observable histories distinguish these channels and whether they help explain addition and replacement separately. No single observed event is assumed to reveal a latent Need state.

**RQ2: Ability—which transactions can the household undertake?** The first subquestion concerns initial funds, including liquid balances and net trade-in equity. The second concerns recurring obligations and the consumption that remains after a transaction. The third concerns available vehicles and contract terms, and how saving or waiting changes feasibility. The question is transaction-specific: does a given purchase leave a feasible consumption and payment plan? Loan approval, willingness to borrow, and the attractiveness of an approved contract must remain distinct.

**RQ3: Choice—which feasible action is preferred, and when?** The first subquestion compares continuation, repair, other transport arrangements, and fleet adjustment. Conditional on adjustment, the next question distinguishes addition, replacement, and disposal. Conditional on purchase, the household chooses vehicle attributes, new or used status, and price tier. Across these alternatives it also compares acting now with waiting. Whether these margins require correlated choice shocks or heterogeneous preferences is a modeling question; the hierarchy in Figure 1 does not itself select a nested-logit specification.

**RQ4: Integration—what can the observations distinguish?** The same income change can affect transport preferences, feasibility, and the value of waiting. The research must determine whether the available data separate these mechanisms, or support only prediction of their combined outcome. A further test is whether the utility formulation improves calibrated purchase forecasts over a simpler discrete-time event-history model using the same information. Predictive improvement, identification of mechanisms, and the causal effect of contacting a customer are separate claims.

### 3.3 State, budget, and feasible actions

To connect these questions to (3), a candidate state is

$$s_{hr} = (\mathcal{X}_{hr}, D_{hr}, \mathcal{L}_{hr}, y_{hr}, m_{hr}, \ell_{hr}, M_r), \tag{5}$$

where $\mathcal{X}$ records the fleet and its relevant attributes, $D$ liquid funds, $\mathcal{L}$ outstanding obligations and their remaining payment schedules, $y$ current income, and $M$ vehicle prices and financing conditions. Stable preference differences can be represented by a household type; any distribution over such types remains to be specified. A state is sufficient only if, together with the chosen action and consumption, it captures the information needed to model subsequent transitions. Vehicle count and oldest age alone need not satisfy this requirement.

Within decision month $r$, the household observes the state and choice shocks, selects $d$ and $c$, meets current outlays, and carries the remaining funds into the next month. With interest credited before the decision, one consistent accounting convention is

$$D_{h,r+1} = (1 + r_r^D)D_{hr} + y_{hr} - c_{hr} - o(\mathcal{X}_{hr}^d, d) - \pi(\mathcal{L}_{hr}, d) - K(s_{hr}, d), \tag{6}$$

where $o$ is current transport operating expenditure, $\pi$ is the payment due this month after accounting for the action, and $K$ is its initial net cash outlay. The timing of the first new-loan payment must

be part of the contract convention. Current payments and initial outlays must not count the same cash flow twice.

For a purchase priced at $P_j$ under a fixed contract with down-payment share $\delta_j$, an illustrative outlay is $K = \delta_j P_j + \text{fees} - E^{\text{trade}}$. Net trade-in equity $E^{\text{trade}}$ equals sale proceeds less debt discharged on that vehicle; it is zero for an addition without a trade-in and may be negative. The new principal and remaining payment schedule enter $\mathcal{L}_{h,r+1}$, while a discharged loan leaves that state. A cash purchase corresponds to $\delta_j = 1$ and no new principal. Treatment of negative equity, fees, and borrowing limits must be made explicit for the contracts actually modeled.

For the initial formulation, financing terms are externally specified by vehicle or price tier, as in the existing working draft. They affect feasibility but are not selected through a lender-choice model. Forecasts under a single standard-contract schedule are conditional on that scenario; representing the population requires evidence about the cash and credit options households actually face.

A simple starting feasible set imposes $c_{hr} \geq c_{\min} > 0$, $D_{h,r+1} \geq 0$, and the applicable contract restrictions. The consumption floor and nonnegative-balance condition are proposed assumptions, not empirically established limits for every household. The initial model must restrict attention to states and transitions admitting at least one feasible action; otherwise an explicit borrowing, payment-distress, or other fallback action is required. Consumption is optimized within the remaining feasible set in (3). Fleet aging, income changes, household transitions, prices, and debt amortization then determine

$$s_{h,r+1} = G(s_{hr}, d_{hr}, c_{hr}, \omega_{h,r+1}), \tag{7}$$

with a transition law for innovations $\omega$ to be specified. Future covariates must be integrated over that law when forecasting from $t$.

### 3.4 From utility to monthly and horizon probabilities

A tractable starting assumption is that the additive shocks are independent type-I extreme value with unit scale, conditional on the modeled state and household type. Under this assumption,

$$\bar{V}_h(s) = \log \sum_{d \in \mathcal{D}(s)} \exp v_h(s,d) + \text{constant}, \tag{8}$$

$$P_h(d \mid s) = \frac{\exp v_h(s,d)}{\sum_{b \in \mathcal{D}(s)} \exp v_h(s,b)}. \tag{9}$$

The shock scale fixes the utility scale; a reference utility normalization is also needed for estimation. These assumptions impose substitution restrictions and would need comparison with alternatives if the data justify them (Train, 2009). Let $\mathcal{D}_P(s)$ collect addition and replacement purchases. The monthly purchase probability is

$$p_h^P(s) = \sum_{d \in \mathcal{D}_P(s)} P_h(d \mid s). \tag{10}$$

For exactly two feasible actions, buy and keep, this reduces to the logistic probability $\Lambda\{v_h(s,\text{buy}) - v_h(s,\text{keep})\}$. Multiple vehicle alternatives require the sum in (10); they cannot be replaced by an unspecified representative car without an additional aggregation assumption.

The horizon calculation must allow nonpurchase actions to change the state. Let $q_m(s)$ be the probability of no purchase over the next $m$ decisions starting in pre-decision state $s$, with $q_0(s) = 1$. Under the candidate model,

$$q_m(s) = \sum_{d \in \mathcal{D}(s) \setminus \mathcal{D}_P(s)} P_h(d \mid s)\, \mathbb{E}\left[q_{m-1}(s') \mid s, d, c_h^*(s,d)\right]. \tag{11}$$

Here $c_h^*(s,d)$ is the optimal conditional consumption from (3); dependence of $q_m$ on household type is suppressed. The end-of-month forecast is then

$$F_h(t,H) = 1 - \mathbb{E}[q_H(s_{h,t+1}) \mid \mathcal{I}_{ht}]\,. \tag{12}$$

This expectation includes uncertainty about the next pre-decision state and, where relevant, unobserved household type. Disposal remains a nonpurchase transition, so disposal followed by purchase is included. The model yields coherent horizons with $F_h(t,12) \geq F_h(t,6)$.

Equivalently, define the forecast-origin hazard

$$\lambda_h^P(r;t) = \mathbb{P}(T_h^P = t+r \mid T_h^P > t+r-1, \mathcal{I}_{ht}).$$

Then $F_h(t,H) = 1 - \prod_{r=1}^{H}\{1 - \lambda_h^P(r;t)\}$. These hazards already average over future histories conditional on remaining purchase-free. Substituting realized future income or life events into that product would change the information set and would not produce a forecast available at $t$.

### 3.5 What must be established before estimation

The formulation specifies how a candidate model could produce the target. It does not yet show that its preference parameters, transition law, or latent components can be recovered from available observations. Normalizations, observed choice sets, variation in vehicle and financing terms, and the initial distribution of states all affect identification. For an initial implementation, the discount factor would be fixed and varied in sensitivity analysis; estimating it requires additional justified restrictions (Magnac and Thesmar, 2002). Scheduled loan payoff, lease expiration, repair spending, and income changes are not automatically exclusions affecting only one layer.

If vehicle condition or another relevant state is unobserved, a state-space extension would require both a transition equation and a measurement equation linking that state to observed signals. Filtering would then condition on information available by the forecast date. Naming a latent state "Need" does not validate that interpretation, and repeated observations alone do not separate stable household preferences from changing conditions. State-space methods provide tools for this extension, not evidence that it is required or identified here (Durbin and Koopman, 2012).

The immediate empirical requirements are a verified household definition, purchase and nonpurchase histories, fleet information, financial resources, and the terms of feasible alternatives. Evaluation should use later periods and account for repeated household observations, incomplete follow-up, missing transactions, and changes in market conditions. Calibration, Brier scores, and log loss at the stated horizons should accompany comparisons with a simpler event-history baseline. Simulation can assess recovery under a specified data-generating process; it cannot establish that the process describes real households. Any benefit from customer contact or a loan offer requires a separate intervention design.

For review, the unresolved decisions are the exact purchase endpoint, the available household and fleet linkage, the financing scenario, and whether the first empirical objective is prediction or an identified counterfactual. These decisions determine how much of the candidate structure can be defended. The literature that follows supplies precedents for its components and makes the remaining evidence gaps explicit.

### 3.6 How the questions are nested

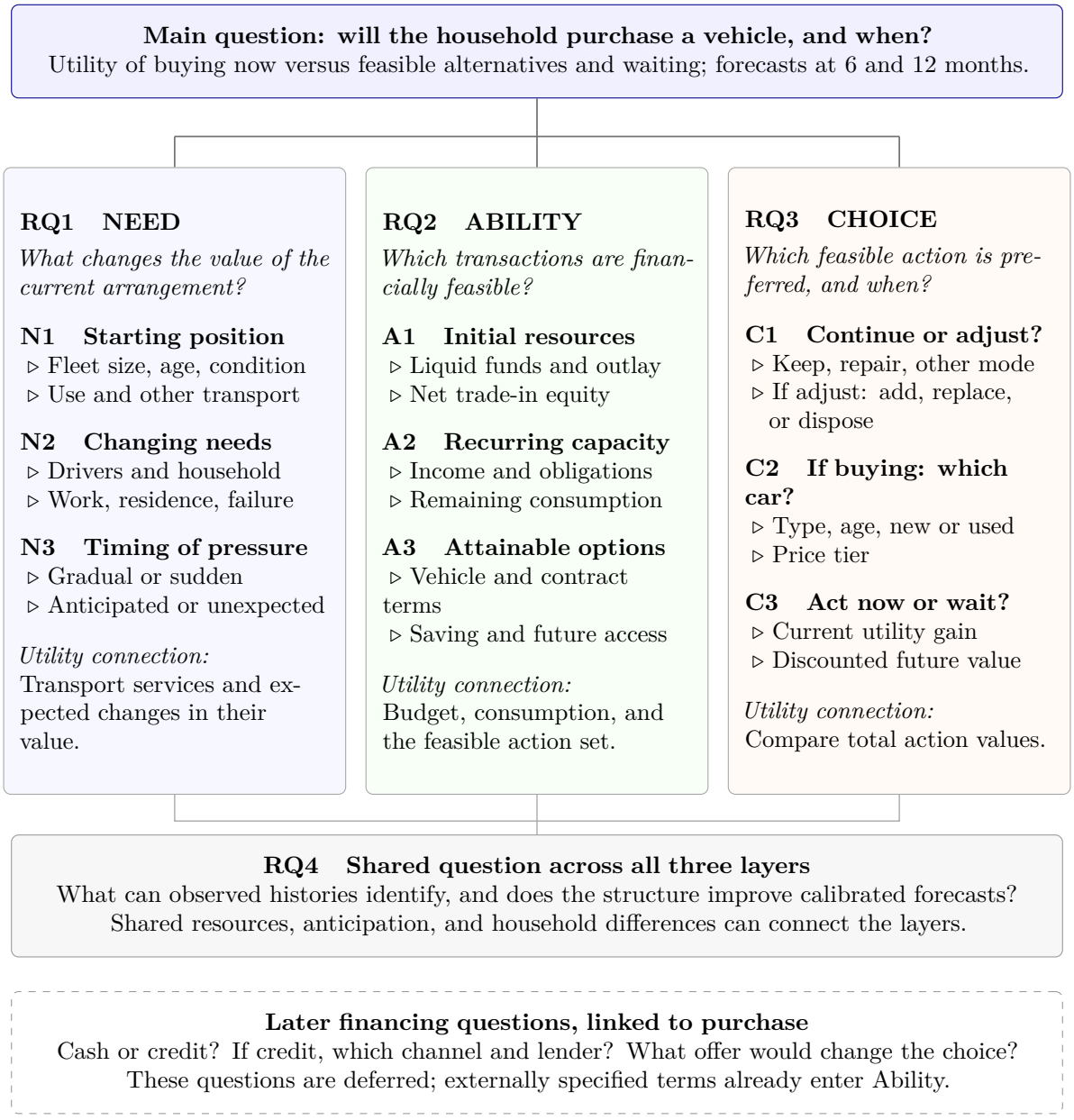



Figure 1: Hierarchy of research questions within Need, Ability, and Choice. The branches show analytical nesting, not a causal sequence, independent gates, or an imposed nested-logit structure. Buying without an identifiable need shock remains possible.

# 4 Model Components and Their Sources

This section states which parts of the formulation in Sections 2 and 3 come from which papers, what we keep, and what we change. Each paper's setup, equations, derivation, and assumptions are retained in the technical appendix of the companion literature review, in the authors' notation; step numbers below refer to that appendix. Papers still pending full-text review are not listed. Which components combine directly and which need added assumptions is summarized in the review, Section 4.

## 4.1 Rust (1987)

*Technical analysis:* literature review, Appendix A.1.1.
**What we keep and what we change.** Keep steps 1–7 with the household in place of the manager: $x_t \to$ fleet state $\mathcal{X}_{hr}$, $RC \to$ net outlay $K(s,d)$, $P(1 \mid x) \to$ the monthly purchase probability (10); our eqs. (8)–(9) are step 6. Change step 4: household data will have persistent heterogeneity, so CI must be relaxed with unobserved types, which then requires an initial-condition term in step 7 (Heckman, below). Add a budget constraint (Gavazza–Lanteri, below).

**Questions for discussion.**

- Which step fails first when a household has a persistent preference we do not observe, and what replaces it?
- What would identify $\beta$ in bank data, or do we fix it and report sensitivity?
- Is a monthly decision period right for vehicle purchase, given that Rust's is monthly but the car papers are annual?

## 4.2 Schiraldi (2011)

*Technical analysis:* literature review, Appendix A.1.2.
**What we keep and what we change.** Keep steps 1–7: $k \to \mathcal{X}_{hr}$ in (5); $\tau_{jt} \to$ fees in $K(s,d)$ and $p_{kt} \to E^{\text{trade}}$ in (6); $\delta_{it} \to$ the purchase log-sum inside (8); (11)–(12) $\to$ (9); $\sum_j d_{it}^{kj} \to$ (10); exit $\to$ a nonpurchase transition in (11). Change step 5: beliefs come from (7). Replace steps 8–10: bank data give household purchase timing, not product shares, so $\tau$ and persistent types must be separated in household panels (Arcidiacono and Miller, 2011; Heckman, 1981). Add several vehicles, the budget in (3), and a monthly $\beta$.

**Questions for discussion.**

- Does Assumption 1 survive once income, debt, and contract terms enter the state?
- Do we fix $\beta$ and report sensitivity, since the recovered $\tau$ moves with $\beta$?

## 4.3 Gavazza and Lanteri (2021)

*Technical analysis:* literature review, Appendix A.1.3.
**What we keep and what we change.** Keep state and budget: $b \to D_{hr}$ ($b$ may be negative; our $D \geq 0$, loans in $\mathcal{L}$), $w \to y_{hr}$, $n \to \mathcal{X}_{hr}$ reduced to one car, $\theta \to$ household type, $(p_b, p_2, p_1, p_N, \phi) \to M_r$ in (5). In (6), $K(s,d) \to [p_{\tilde{n}} - (p_n - \lambda(p_n))]\mathcal{I}(\tilde{n} \neq n)$, a cash purchase with $E^{\text{trade}} = p_n - \lambda(p_n)$,

and $o, \pi \to 0$. Keep the limit as feasibility (step 7), relaxed by resale value via (11). Change: add EV1 shocks (step 8), a monthly period, a loan schedule in $\mathcal{L}$, and estimation on household data. Fix one interest convention: (6) implies $b' = (1+r)[b + w - c - \tau - K]$, interest on closing savings; (6) credits the opening balance. Take prices as given (drop step 10).

Risk attitude is out of our scope, but the credit-shock mechanism runs through precautionary saving (review step 1), so we must take a position on $\gamma$ rather than inherit 2.

**Questions for discussion.**

- With EV1 shocks, does an option failing step 7 leave the choice set or just get low utility?
- Which $\gamma$ do we adopt, and with what sensitivity range?
- Can bank data observe $\phi$ per customer, or must it stay a calibrated common limit?

### 4.4 Magnac and Thesmar (2002)

*Technical analysis:* literature review, Appendix A.2.1.

**What we keep and what we change.** Map $i \to d \in \mathcal{D}(s)$, $K \to$ keep, $x \to s_{hr}$ in (5), $u_i^* + \beta\mathbb{E}(\cdot) \to v_h(s,d)$ in (3), $\tilde{p} \to P_h(d \mid s)$ in (9), $G \to$ the EV1 assumption behind (8), $\theta \to$ household type. Keep steps 2–6 as the list of what we fix: $\beta$, $G$, and the keep utility in *every* state are identifying assumptions, not conventions. Change the habit of setting $u^*_{\text{keep}} = 0$: if keeping costs more as the vehicle ages, that loads the age effect onto the buy utility. *Consequence:* with flexible utilities, these normalizations do not change fitted choice probabilities or transitions, so a forecast of $F_h(t, H)$ in (12) inside the observed environment is unaffected; they do change recovered utilities and therefore counterfactuals (new prices, contract terms, a loan offer).

**Questions for discussion.**

- Is our first objective the in-environment forecast $F_h(t, H)$, which needs no identified utilities, or a counterfactual, which does?
- Which state variable could satisfy $U_i(x_1) = U_i(x_2)$? A scheduled loan payoff shifts the future but also changes $\mathbb{E}(v_K^*(x') \mid x, d = K)$, so it is not automatically an exclusion.
- With sparse household cells, is $\Delta$ far enough from zero to make $\hat{\beta}$ informative, or do we fix $\beta$ and report sensitivity?

### 4.5 Hotz and Miller (1993)

*Technical analysis:* literature review, Appendix A.2.2.

**What we keep and what we change.** Keep steps 2–5 and 7: our (9) is step 3 with $v_h(s,d)$ in place of $u^* + v$, and (8) combined with step 4 writes the continuation value in observed probabilities. Our (11) and (12) already need only $P_h(d \mid s)$ and (7), so a predictive forecast could use first-stage estimates directly. Change step 6: no household action is terminating, and a purchase resets vehicle age but adds a loan, so the renewal representation of Arcidiacono and Miller (2011) must be checked for the full state (5). Change step 7: continuous $D$, $y$, $\mathcal{L}$ in (5) make cell frequencies infeasible, and household types require the extension of Arcidiacono and Miller (2011).

**Questions for discussion.**

- Is the first objective prediction, where first-stage CCPs may suffice, or counterfactual financing terms, which need the structural parameters?
- How coarse a state can we accept for first-stage CCPs before smoothing over balances and income distorts $\ln p$?
- Which household action, if any, is close enough to a renewal to keep the continuation values one period ahead?

### 4.6 Li et al. (2011)

*Technical analysis:* literature review, Appendix A.3.1.
**What we keep and what we change.** Keep steps 3–6: Li's $s$ becomes a latent Need component added to (5), driven in (7) by vehicle age, life events, and past purchases, with (12) as our initial-condition equation; $\mathbb{P}(Y_{ijt} \mid s)$ maps to $p_h^P$ in (10) and $\mathbb{P}_{it}(s)$ to the filtered belief in (12). Change step 1: our choice equation is (9), with the continuation value of (3), not a static probit. Change steps 3–5: use the discrete-time version or add elapsed duration to the state, since the semi-Markov state would break the recursion (11). Add a measurement equation for transaction signals and estimate with Arcidiacono and Miller (2011).

**Questions for discussion.**

- Do balance and transaction variables belong in utility (Ability) or in a signal equation for Need?
- Does bank contact enter our model? If so, what makes it exogenous given the latent state?

### 4.7 Adams et al. (2009)

*Technical analysis:* literature review, Appendix A.4.1.
**What we keep and what we change.** Keep the design: separate the current outlay from deferred payments and test which matters. $d_i$ maps to $\delta_j P_j$ in $K = \delta_j P_j + \text{fees} - E^{\text{trade}}$ in (6), $(p - D)\phi$ to the schedule in $\mathcal{L}_{h,r+1}$, a refund to a predictable jump in $y_{hr}$, and the probit index to $v_h(s, \text{buy}) - v_h(s, \text{keep})$ with normal shocks. Change the outcome: Table 2 is $\mathbb{P}(\text{buy} \mid \text{applied here})$, while (10) and (12) need an unconditional monthly probability with waiting. Do not transfer the subprime magnitudes to prime customers.

**Questions for discussion.**

- Does bank data vary down-payment requirements for prime customers, or must we identify the effect of $K$ from liquid balances $D_{hr}$?
- Is the February spike added purchases or purchases moved earlier, and which matters for a 6- or 12-month horizon?

### 4.8 Heckman (1981)

*Technical analysis:* literature review, Appendix A.5.1.

**What we keep and what we change.** Map $i \to h$, $J \to$ the first month household $h$ appears in bank data, $\tau(i) \to$ the household type, and $d(i, J) \to$ the first observed state $s_{hJ}$ in (5). Keep step 4: our likelihood, built from (9) and (7), must include $\mathbb{P}(s_{hJ} \mid \text{type}, \text{pre-window information})$, since households seen with younger vehicles are disproportionately high-turnover types. Keep step 7 as a warning against household fixed effects with lagged fleet states in short windows. Change step 8: the initial state is multivariate (fleet $\mathcal{X}$, ages, loans $\mathcal{L}$, balances $D$), so a binary probit does not apply; with discrete types, the analog is type probabilities that depend on $s_{hJ}$ and pre-window covariates (Arcidiacono and Miller, 2011). Step 3's stationarity conflicts with trends in $M_r$.

**Questions for discussion.**

- What pre-window information (account history, bureau fields, loan origination dates) do we have for each household, and does it predict $s_{hJ}$?
- Do we model the full joint initial state, or a lower-dimensional summary, and what does that approximation cost in consistency (the p. 189 caveat)?
- Given the 25-replication evidence, should we run our own Monte Carlo at bank-panel sizes before choosing an initial-conditions estimator?

### 4.9 Open questions

- Is the first objective a forecast of $F_h(t, H)$, for which the normalizations above do not affect fit, or a counterfactual response to terms, for which they do?
- Should household Need be a latent Markov state, or are persistent household types sufficient for a first version?
- Is a monthly decision period required?
- How should vehicle purchases not financed through Chase, which the bank may not observe, enter the outcome?

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